

# Forced Smile Creates Positive Mood: A Qualitative Analysis With A VR Experience

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**Figure 1: Left: When the player smiles, their virtual avatar in the mirror reflection—a pink ray fish—simultaneously smiles. Right: The players smile during the VR experience.**

## Abstract

We developed a VR project that uses smile tracking as a novel interaction mechanic to foster emotional well-being and prompt reflection on the meaning of a smile in social connection. This is based on a broad literature that indicates that ‘faking’ a smile can result in a genuine improvement in mood. We conducted an experiment in 60 participants within two between-groups and analyzed the qualitative data of 11 selected participants in this paper in order to understand in detail what the experience of faking smiles is like. Through guided exploration in a VR environment and smile-triggered transformations of NPCs, players reported improved mood, relaxation, and a sense of healing. Although some initially perceived the mechanic of the smile as ‘forced’, they ultimately transitioned to genuine amusement and deeper engagement. In addition, the experience highlighted the participants’ realization

that their real-world smiles are sometimes too subtle, potentially affecting social bonds.

## CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Empirical studies in interaction design**; **Interaction techniques**; • **Applied computing** → **Media arts**.

## Keywords

Virtual Reality, Smile Interaction, Social Connection, Qualitative Research, Affective Design, Emotional Well-being,

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## 1 Introduction

In an era marked by growing social isolation, Human-Computer Interaction (HCI) scholars are increasingly examining how immersive technologies might foster connection and positive affect. Virtual

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reality (VR) enables users to experience new forms of embodiment and emotional involvement, yet few VR systems directly leverage the user's own facial expressions to induce mood changes. Drawing from the hypothesis of facial feedback[3][16], we hypothesize that repeated smiling, even if initially prompted, can become genuinely uplifting over time.

This paper introduces our project, an interactive virtual reality experience in which participants explore an underwater world by gently sweeping their arms, gliding toward ghostly fish, and smiling to bring these creatures to life. Through our novel smile-tracking mechanic, the color palette and ambiance shift from cold blues to warm, golden hues, illustrating a metaphor of emerging from isolation. We propose that this approach not only heightens immersion, but also offers users a tangible sense of agency over their emotional state. In this paper we present a qualitative analysis of user interviews that does not simply ask whether forced smiles improve mood, but what is the detailed experience of forced smiles and what other features of the environment can reinforce the effect.

Specifically, we ask: **(RQ1)** How do smile-driven interactions in VR influence users' emotional engagement and perceived social connection? **(RQ2)** How do users' body movements and facial expressions enhance or disrupt their sense of being immersed in the VR experience? **(RQ3)** How can this VR experience guide designers to create VR environments that help participants move from intentionally smiling to genuinely feeling happy?

## 2 Related work

VR can provide aesthetic experiences that effectively integrate health education, induce healthy emotional states, and help people reflect on their own healthy self-reflection. VR-based art therapy is an innovative approach with great potential. However, more empirical research is needed to confirm its effectiveness in supporting mental health, encouraging emotional release, and promoting positive health behaviors [14].

In terms of using VR to induce positive emotions in people, Collange and Guegan [4] shown that VR is a promising method to induce gratitude, with implications for improving well-being and social connections. Felnhofer et al. [6] explore the efficacy of virtual environments as mood-induction procedures, conducting studies with 120 students in the five predefined virtual park scenarios; this study found that almost all virtual park scenarios successfully elicited the intended emotions: joy, sadness, boredom, anger, and anxiety, and presence levels were consistent across all scenarios.

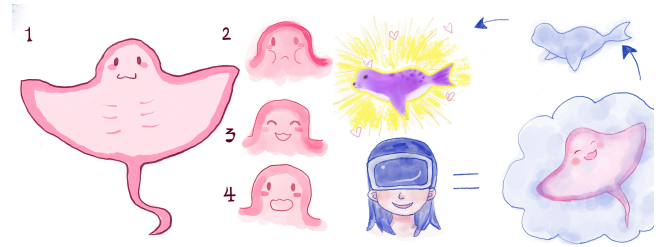
According to James-Lange theory of emotion [3], emotions arise from the perception of physiological responses to stimuli. Previous studies have shown that adopting pseudo-facial expressions, such as simulated smiles, can positively influence emotional and physiological states; specifically, the experiment by Strack, Martin and Stepper [16] was a cornerstone of research into the facial feedback hypothesis, which showed that participants who held a pen in their mouth (forcing a smile) reported more positive emotions than those who did not. Kraft and Pressman [7] have shown that fake smiles can help in stress recovery by influencing both emotional and physiological states.

In recent years, researchers have started to use technology to help people practice smiling. Tsujita and Rekimoto [18, 19] developed a device 'HappinessCounter' to make people recognize their mood and force them to smile, and participants reported a large improvement in mood after the experiment. Sae-Lao et al. [12] developed an endless-runner game with a smile interface, and the results show that the input of smiles as a game mechanism leads to more enjoyable gameplay than other input methods, such as without smile and boosting or auto-boosting without smile. In Tranon et al.'s experiment [17], there was no significant difference in positive and negative effects when using facial expressions (i.e. smiling or frowning) or a controller button as an input mechanism in an HMD-based AR game. We noticed that there are few researches focused on using smile as a game mechanism in VR, therefore, we want to fill the gaps in this area.

These results show that using smiles as an interaction mechanic can induce positive mood, but the quantitative results above do little to explain what the experience of these forced smiles is like. What does it feel like to fake a smile? The results also do not fully explore the impact of other features of an experience on fake smiles. In this paper we present a qualitative analysis of our VR experience that probes these experiential issues.

## 3 Implementation

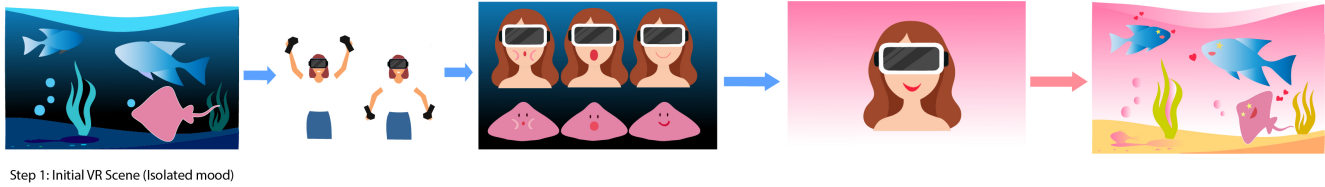
### 3.1 Concept Design



**Figure 2: (Left) Design sketches of various facial expressions for the rayfish. (Right) Mechanism sketches illustrating how smiles trigger NPCs to transform from ghosts state to realface state.**

We developed a novel smile-tracking mechanic that directly translates the player's smiles into environmental changes. The player awakens in a cold virtual ocean, and they first see their reflection, a pink ray fish, inside a bubble with a mirror-like surface. They can blink at the bubble mirror, puff out their cheeks, swing their body, smile, or open their mouth, and the mirrored image immediately mirrors these actions.

A gentle jellyfish guides players through the ocean by first demonstrating how to use VR controllers with wide arm movements, similar to embracing the world, to swim forward. Each time the players follow the instructions of the jellyfish and reach a new goal, they receive positive visual and auditory feedback. This feedback creates a sense of achievement through environmental cues. We designed an interaction that allows players to slide their arms smoothly, simulating swimming movements. When players successfully move their arms, the vibration feedback from the VR



**Figure 3: Step 1. Initial VR Scene - Player takes on the role of a pink ray fish swimming in the blue ocean. Meet the semi-transparent, blue ghostly fish (NPCs); Step 2. Swim with arms to move; Step 3. Facial tracking-Detecting facial expression via the Meta Quest Pro headset; Step 4. Smile Sustained  $\geq 10$ s; Step 5. Trigger environmental change-Fish transforms to real form; Environment shifts from cold blue to warm pink hues; Heart-shaped bubbles appear.**

controllers emulates water ripples, accompanied by water wave sound effects, thereby providing players with the sensation of genuinely swimming in water. In order to enhance the player's sense of embodiment [15], these designs create synchronizing visual and tactile feedback, allowing the player to have convincing illusions of body ownership, which are crucial for enhancing user engagement and emotional connection within virtual settings.

Continuing alongside the jellyfish, the player arrives at another pearl with a reflective surface. Subsequently, the jellyfish uses friendly and gentle language to remind players to smile. The moment the player attempts to smile, the entire ocean shifts from cold blue to warm yellow and pink. At the same time, they noticed numerous translucent ghost-like sea creatures in the water. When the player swims closer and smiles at these creatures for more than ten seconds, the creature responds to each new smile by emitting a glow, and gradually shedding their spectral forms and becoming vibrantly alive. Heart-shaped bubbles drift upward, weaving a sense of connection. We hope that this design demonstrates that seeing and acknowledging others gently builds understanding. There is background music with slow and expansive rhythms that parallel the player's journey, evoked a calm dreamlike state.

### 3.2 Technical Implementation

We developed this VR project in Unity (2022 3.11 f2), using C-sharp scripts to track and respond to user facial movements through the built-in face tracking API of the Meta Quest Pro VR headset, which provides real-time data on facial expressions. We integrate Oculus Integration SDK (v57.0) to create interactions and use Meta Movement SDK (v1.2.0)[8] to detect when the user holds a sustained smile above a defined intensity threshold. We tracked the left and right lip corner puller, when the threshold of lip corner is  $> 0.3$ , we define it as smile; when the threshold of lip corner is  $< 0.1$ , we set it as end of smile. Once activated, the VR environment responds by altering underwater lighting, transforming ghostly fish, and logging event data (e.g., smile duration, frequency). The VR environment renders a stylized underwater world within Universal Render Pipeline (URP) 14.0.11 with assets assembled from Low Poly Underwater Pack v1.2.0 [20].

## 4 Methodology

We conducted an empirical study. In an effort to explore how smile-driven interactions in VR influence both emotional and experiential outcomes, we recruited 60 participants, each participant engaged

in a VR project in a one-time session. We used between-subjects design to divide 60 participants into two different groups: Smile group and Open-Mouth group, however in this paper we will only analyse qualitative data from the Smile group. During their session, participants wear the Meta Quest Pro VR headset equipped with facial tracking technique. During the VR session, we used the Meta Quest Pro VR headset to record people's facial data. After the VR session, we collected both quantitative and qualitative data. After the full experiment session, all participants each received a reward of £15 Amazon voucher.

### 4.1 Participants

We recruited participants on the campus of the study university. We invited potential participants to complete an online UCLA Loneliness Scale (version 3) [11]. A total of 60 participants with loneliness score no smaller than 31 were selected. The median of their UCLA Loneliness Scale score is 48: with points lower than 48, we define it as the lowest and medium degree of loneliness level; with points higher than 48, we define it as the medium and higher degree of loneliness level. Among them, there are 39 women and 21 men, 24 participants aged between 18-24, 32 of them aged between 25-34, and 4 are aged between 35-44. In this paper, we focus on the SMILE group (15 participants  $< 48$  score: mean = 40.06; 15 participants  $\geq 48$ : mean = 55.06). Participants were fully informed about the specific nature of the research question prior to the experiment.

### 4.2 Data Collection

Participants completed the loneliness state scale during recruitment. Before the VR experience, participants completed a pre-questionnaire that integrates the PANAS schedule [21], the Big Five personality [5], the Self-Compassion Scale [9]. These surveys were designed to capture subjective data on the participants' psychological states and perceptions. Immediately after the VR session, players were invited to respond to a post-experiment questionnaire. This questionnaire consists of the PANAS Schedule [21] UCLA loneliness scale [11], self-compassion scale [9] and the flow state scale [10]. After the post-experiment questionnaire, each participant was interviewed for approximately 15 minutes to explore their subjective experiences. Participants are invited to respond to nine open-ended questions. After two weeks, participants were invited to evaluate the long-term effect of this VR experience by completing a follow-up survey using the UCLA Loneliness Scale [11] and the self-compassion scale [9]. They responded to one open

| ID  | Gender | Age   | Nationality          | Change in Loneliness | Change in Positive Emotion | Change in Negative Emotion |
|-----|--------|-------|----------------------|----------------------|----------------------------|----------------------------|
| P1  | F      | 25-34 | Chinese              | -1                   | 5                          | -6                         |
| P2  | F      | 25-34 | Chinese              | -2                   | 8                          | -11                        |
| P3  | F      | 25-34 | Chinese              | -5                   | 5                          | -9                         |
| P4  | F      | 25-34 | Chinese              | -1                   | 5                          | -12                        |
| P5  | F      | 18-24 | Chinese              | -4                   | 6                          | -6                         |
| P6  | M      | 35-44 | Egyptian             | -5                   | -4                         | -7                         |
| P7  | M      | 25-34 | French and Italian   | 11                   | 17                         | -16                        |
| P8  | F      | 25-34 | American and British | -3                   | -1                         | -9                         |
| P9  | F      | 25-34 | Chinese              | -1                   | 6                          | -1                         |
| P10 | M      | 25-34 | British              | -1                   | -1                         | -19                        |
| P11 | F      | 25-34 | Chinese              | -10                  | 5                          | -1                         |

Table 1: Participant's details of this qualitative study

question: *What impact, if any, do you think this VR experience might have on your real-life social interactions or emotional well-being?*

This paper will focus on the qualitative analysis of the interview data and the open question from the follow-up survey.

## 5 Qualitative analysis

We transcribed all interviews, but ultimately selected data from 11 participants in the SMILE group for thematic analysis [2] in this paper. The selection was guided by quantitative results: we included 11 participants whose negative emotions scores decreased, including 8 participants whose positive emotion scores increased and 3 participants whose positive emotion scores decreased. The details of the demographics of the participants are shown in Table 1. In this phase of the study, we focus solely on the qualitative data of these 11 participants. All participants are invited to respond to nine open questions:

- How did you feel during the VR experience? Please describe your emotions and any specific moments that stood out to you.
- Please describe if there are any thoughts that appear when you experience this project? What are you thinking about? Or you are only focusing on the experience.
- How did you feel about other fish and sea creatures at the very beginning?
- How did you feel about the sea creatures change over time?
- How did the use of facial expressions make you feel?
- How did swimming with your arms make you feel?
- What impact, if any, do you think the VR experience might have on your real-life social interactions or emotional well-being?
- Were there any parts of the VR experience that you found challenging or uncomfortable? Please explain.
- Is there anything else you would like to share about your experience in the study? Or to recommend to us?

### 5.1 Theme 1: The mechanics of smiling: Forced vs. Spontaneous expression

The participants feel their self-consciousness—shifting from a mechanical, forced, and goal-oriented game action (*"I must smile at fish to change them"*) to a surprising sense of real, internal positivity. During this process, we find psychological and physiological feedback loops. As P1 said, *"It starts as fake laughter but eventually becomes real laughter."* At first, some participants feel forced to smile, as P2 states, *"At first, I felt a bit stiff because I knew it was a programmed mechanism. Smiling felt a bit forced and awkward. Initially, it felt a bit like I was just stretching my mouth awkwardly."* But as time passed, P2 said, *"I started genuinely smiling and then it felt good."* The feeling towards smiling also depends on the extraversion level. In addition to that, the mechanics of forced smiling can help introverts stretch themselves and get a natural and comfortable smile. As P3 mentioned, *"At first, my smile was small because I am naturally more introverted. My faint smile was not easily recognized, so I had to put on a proper smile, which I thought was a clever design, it encouraged me to smile genuinely. This immediately made me feel relaxed and at ease."* Such forced smiling leads to genuine emotional engagement, we finally find how "pretend" smiling leads to elevated mood. This transition from a "fake" smile to a real smile indicates a moment of self-awareness and engagement with the mechanics of the VR experience, which led to a more authentic emotional response. It shows that even when a smile is initially performed as a mechanical input, the immersive qualities of VR can transform that forced gesture into genuine positivity, suggesting that well-designed facial interaction loops can foster authentic emotional change.

### 5.2 Theme 2: Elements make people feel healing

**5.2.1 Comfort and safety.** Participants described a profound comfort in the undersea environment. As P1 explained, *"I really like the feeling of being surrounded by water, it gives me a sense of security. It felt very calm and peaceful."* Meanwhile, P3 highlighted the impact of seeing one's own smile reflect back, *"Seeing my reflection smiling back felt reassuring."* These remarks underscore how both ambient underwater design and personal affirmation through the mirror can nurture a higher sense of emotional security. Many participants

reported an immediate boost in mood after interacting with the VR environment. *“About 30 seconds in, I felt much happier”* (P1), a user noted, underscoring the rapid onset of positivity. *“I thought it was quite amusing”* (P4) similarly highlighted the fun atmosphere of the experience, while P5 tied these feelings to the transformation of the sea creature: *“I felt happy when they transformed into their real forms because they smiled back at me. That moment of feedback made me happy.”* P9 liked this transformation moment as well, because *“that moment gave me a warm and uplifting feeling.”* Another user, P11, mentioned that it was *“the time I smiled the most recently”* and concluded, *“It left me feeling very sunny and positive”*. The VR experience had a clear positive effect on the participant’s immediate emotional state. Even though they are unsure about long-term or subconscious influences, they report feeling tangibly happier right after the session, supporting the notion that smiling in VR can boost short-term mood.

**5.2.2 Humor and Amusement.** Several participants reported humor that arises from both the forced nature of smiling and the whimsical design of the VR space. *“I thought it was quite amusing,”* P4 explained, likening the experience to *“fake socializing... like taking staged photos... I felt like the boy in that meme with a forced grin”* [13]. However, this very fakeness led P4 to *“laugh for real,”* revealing an unexpected change from forced grin to genuine amusement. P4 also found humor in taking on the role of *“a fun-looking stingray with silly expressions”*. In contrast, P5 described the delight of *“watching fish transform into their real forms”*, while noting that *“searching for them... felt like something [they] had not tried before”*, thus underscoring how novel mechanics and playful visuals combined to spark a sense of fun.

**5.2.3 Escape and Focus.** Many participants recounted how the VR setting temporarily freed them from daily stress. *“Before entering, I was very anxious,”* P1 shared, *“but I didn’t have time to think about real-life concerns... This experience helped me focus on the present moment.”* Similarly, P5 noted being *“entirely focused on the experience”* without external associations, and P7 described the same singular attention. P11 elaborated on a *“wordless, language-free state”*, saying that it was only in reflection after the experience that they realized that they had reached such deep immersion. These accounts underscore the ability of the virtual reality environment to provide mental respite and encourage users to be fully engaged in the immediate moment.

**5.2.4 Curiosity and Exploration.** This VR experience sparked curiosity and amusement. Many participants approached the VR environment as a space for playful testing and discovery. P8 described *“getting used to the world and how to explore it,”* eventually becoming eager to *“do weird things with my face... and see what happens.”* This sense of experimentation was echoed by P9: *“It caught my attention... They [the sea creatures] became individuals that I wanted to identify.”* P10 similarly demonstrated a testing mindset, *“experimenting with if smiling more would make the sea creatures appear more quickly,”* showing how participants probed the rules of the system to satisfy their curiosity and gain a greater sense of control.

**5.2.5 Visual and Aesthetic Shifts: Color Transitions and Emotional Tone.** The calm music and soft colors provide a relaxing environment for the player. The role of changing color palettes (e.g., deep

blue to golden pink) and overall environmental design (e.g., golden mountains) evoked emotional warmth. To create an empowering VR experience, visual aesthetics and auditory design can significantly impact emotional responses, bridging participants’ subjective experiences with the crafted elements of the VR environment. It underscores VR’s potential for multisensory emotional design. From auditory pleasure, as P9 states, *“The sound design was great—whenever I swam, the sound of water made it feel very realistic, like I was truly swimming.”* The visual design, particularly the use of golden tones, has a significant emotional impact, which causes participants to associate these warm golden visuals with feelings of comfort and positivity. As P1 said, *“I also liked the environment; it felt very calm and peaceful. This kind of environment helped improve my mood [...] I especially loved the color change—it felt like the transition from a cold tone to a warm tone. The color interaction felt more direct and impactful. The tonal shift brought me a strong visual experience.”* P9 also suggests that aesthetic choices in the VR environment play an important role in improving emotional engagement, saying that *“after smiling long enough, the transition from blue to gold and then pink was incredibly warm and heartwarming to me. I also noticed that when I smiled for the first time, the ocean environment changed from a cold blue tone to a warm yellow tone. It felt very warm and comforting. [...] Your visual design reminded me of that, dreamy and beautiful, I also loved the golden mountains in the scene, it gave me a warm feeling. It did not have that cold and desolate vibe that some underwater scenes have”*.

### 5.3 Theme 3: Educational value: From ‘Ghostly’ to ‘real’ metaphors of social connection, how the force smile generates people’s reflection about social connection

In this experience, we set the required smile duration to 10 seconds. Only when the player has continuously smiled at the fish for that full duration does the fish respond and reveal its true form. This makes participants associate those socially awkward moments with real life. As presented in P3, *“At some points, I smiled multiple times at a fish, but it did not react. That felt awkward, like giving warmth and not getting anything back. It reminded me of social interactions where enthusiasm is not reciprocated. Smiling feels like an expression of warmth and friendliness, so when there’s no immediate feedback, it creates a sense of disappointment.”* P6 believes this situation makes sense, he said, *“because not all people are the same, some people need you to put more effort into it, it’s like in the real world.”*

In another aspect, this VR experience mechanics elicits some participants’ reflection about social behaviors. For example, P5 believes that the VR experience has the potential to promote positive emotional states and social behaviors, such as patience. As P5 said, *“I wondered if my smiles were too subtle because there were many times when I smiled for a long time, but they did not respond. It might remind people to be more patient and maintain positive emotions for longer periods.”* In addition to this, the participant experienced strong positive emotions when the fish transformed after they smiled at them. Feedback from virtual creatures, particularly when they smiled, created an emotional connection and a sense of joy. As P5 said, *“I felt happy when they transformed into their real forms because they smiled back at me, that moment of feedback made me happy. I*

found it exciting to wait for their transformation.” This suggests that the act of receiving feedback in the form of transformation was a key emotional trigger in the experience.

The act of smiling at the virtual creatures and witnessing their transformation fostered a strong sense of emotional connection. As P11 said, *“Emotionally, I noticed a fish that was initially transparent. After I smiled at it, it became solid and I could see its face. It reminded me of real life... if you take the first step to show friendliness, the surroundings often give you positive feedback [...] It's like doing something positive and receiving a good response in return. It made me realize that my smile could bring warmth to others.”* This participant draws parallels between the VR interaction and real-world social behavior, likening the transformation of the sea creature to the way people respond positively to warmth and friendliness. This highlights how VR can simulate or reinforce social reciprocity in a playful and tangible way. Furthermore, the participant's awareness of genuine versus forced smiles leads to a more broader reflection on emotional expression beyond the VR setting.

In addition, the participant becomes more aware of how their facial expressions register, both in real life and within the VR system. As P9 mentioned, *“One thing that made me realize is how much effort it takes to truly smile. During the experiment, I observed moments in which the system did not respond because my smile was not strong enough. It might help people realize that they need to smile more visibly.”* P11 also said, *“It encouraged me to open up genuinely and smile more freely.”* These insights suggest that individuals may often undercommunicate warmth or friendliness if their facial signals are too faint. In everyday interactions, nonverbal signals, such as smiling, play a key role in conveying approachability and building rapport. If a smile goes unnoticed or appears insincere, it can hinder genuine connection. Recognizing how facial expressions register (or fail to register) in a digital environment can translate to real-life awareness: People may become more intentional about expressing clear and confident non-verbal cues, ultimately fostering stronger social bonds and better mutual understanding.

#### 5.4 Theme 4: Culture shock may cause different attitude and level of face muscle fatigue in the smile

The different feelings and attitudes towards being forced to smile for 10 seconds to change the state of sea creatures probably depend on different national cultures. Three participants from China, where people often suppress or moderate their emotional expressions to maintain social harmony[1], described experiencing noticeable fatigue in their facial muscles after the experiment. P4 said, *“I found it a bit tiring - until the end, it made my face feel stiff.”* P5 reported, *“The 10-second smile requirement was a bit challenging. —it felt forced.”* P6 said, *“It felt like the signal was, I need to give warmth to others for them to respond.”* Rather than smile at another object in the world, P6 prefers *“If the experience aims to encourage people to smile and feel better, I'd rather smile in the mirror, engage with my own emotions than chase fish and smile at them.”* This reaction may be linked to cultural norms that emphasize emotional restraint, making prolonged or exaggerated facial expressions feel unnatural.

According to self-reported information, things are different for those participants who come from Egypt (Arab), British (Black),

Italy (half Italian, half French). P7 states *“When I smiled, I didn't feel tired, [...], I am not uncomfortable.”* P8 said *“I don't feel tired, [...], most Italians like to smile.”* P9 has some reflections about the cultural difference: *“I used to smile before coming to London. But here is like saying, smiling is not big, when you smile at some strangers.”*

## 6 Discussion

In our preliminary qualitative analysis among 11 participants of varied cultural backgrounds, our findings add empirical support to facial-feedback theories, suggesting that even simulated smiles can evolve into genuine positive feelings under immersive conditions. The findings address the three main research questions as follows:

- RQ1: How does focusing on smiles in VR affect participants' feelings of happiness and social connection?

In our VR experience, although participants initially reported a sense of self-consciousness, feeling that their smiles were in a mechanical response to in-game prompts, however, as the experience progressed, these forced smiles transitioned into genuine amusement and warmth. This shift was accompanied by immediate mood enhancements, such as feelings of warmth, relaxation, and increased positivity. The act of smiling led to tangible transformations within the virtual environment, such as cold–warm color transitions, the symbolic transformation of virtual creatures (from ghostly figures to vibrant life forms), and multisensory feedback. The VR environment effectively communicates messages that participants can interpret and integrate. In doing so, they discovered the underlying metaphors and drew connections between interactive behaviors, immersive content, and triggered events in VR and their own real-world social interactions, which encouraged their self-reflection. This result meets our core design objective. Through smiling in a well-designed VR experience, participants experience a restructuring of their emotional state and engage in a meaning-making process.

- RQ2: How do users' body movements and facial expressions enhance or disrupt their sense of immersion in the VR experience?

The integration of bodily movements and facial expressions played a crucial role in cultivating a strong sense of presence within the VR experience. The participants engaged in graceful swimming motions and deliberate smiling, which were mirrored by their avatars and directly influenced the virtual environment. This embodiment fostered a heightened sense of immersion, making users feel as if they were genuinely part of the underwater world. Synchronized movements and expressions created a seamless interaction between the user and the VR environment, enhancing the illusion of being “inside” the experience. However, cultural differences and varying comfort levels with facial expressions occasionally disrupted this sense of presence for some participants, particularly those from cultures where expressive smiling is less common. Despite these disruptions, the overall design effectively uses bodily and facial interactions to deepen immersion and emotional engagement.

- RQ3: How can this VR experience guide designers to create VR environments that help participants move

from intentionally smiling to genuinely feeling happy?

From these observations, designers can draw several key insights: **Responsive feedback loops:** Implementing mechanics where user expressions directly influence the virtual environment can create meaningful interactions that foster emotional responses. **Embodiment and Presence:** Seamlessly integrating and mirroring user's bodily movements and facial expressions through avatars can strengthen the sense of presence and enhance emotional engagement within the VR experience. **Gradual Transition Mechanisms:** Designing experiences that allow users to transition from intentional to authentic expressions can facilitate deeper emotional connections and sustained positive outcomes. **Metaphorical Storytelling:** Designers can use metaphors within the narrative framework, combined with interactive mechanics, to enhance emotional engagement and provide meaningful context for user actions. This approach can deepen the emotional impact of interactions and promote a deeper sense of connection and well-being.

## 6.1 Limitation and Future work

First, our current findings are based on self-reported data, which can be subject to response bias. The responses provided by the participants may be consistent with perceived expectations rather than with their actual emotional state. To address this issue, future studies could integrate physiological data to provide more objective measures of emotional responses. Combining self-reported experiences, behavioral interactions, and physiological signals could provide a more comprehensive understanding of how VR-based smile interactions affect emotions and social connections.

Second, our results suggest that forced smiles in VR can enhance mood. In future iterations, we aim to explore how to encourage spontaneous smiles rather than relying on prompted or forced ones. A key question for future research is: How can VR interactions be designed to elicit genuine, natural smiles, and how do these compare to intentional smiles in terms of emotional impact? The research in this difference could provide more insights into the authenticity of emotional engagement in virtual environments.

Additionally, the VR storytelling and interaction mechanics could be further refined to enhance player engagement and emotional authenticity. We need to ask how to reinforce emotional shifts more organically, in order to strengthen immersion and long-term emotional benefits.

## 7 Conclusion

Our research demonstrates that smile-driven interactions in VR can transform forced mechanical expressions into genuine positive emotions. Our qualitative analysis shows that immersive VR environments, designed with interactive facial feedback loops, can foster emotional transformation, enhance social connections, and improve well-being. We have demonstrated that integrating body movements and facial expressions into responsive environments could enhance presence and emotional engagement.

Our findings reinforced psychological theories linking physical expressions with emotions and highlighted VR's potential to promote emotional well-being through embodied interactions and

metaphor-driven storytelling. The shift from forced smiles to genuine pleasure suggests that interactive design can naturally shape user emotions.

Our work also highlighted that cultural and personal differences could influence experiences with facial interaction mechanics. Although some users found the mechanics of smiles engaging, others, particularly those of cultures where emotional expression is often moderated or suppressed to maintain social harmony[1] - experienced discomfort or facial fatigue. These insights inform future VR design, guiding the development of more inclusive and adaptive environments.

Future research should examine the long-term effects of smile-driven interactions and explore ways to encourage spontaneous smiles. Incorporating physiological and behavioral data could deepen the understanding of emotional impact. Exploring VR's role in therapy, social VR, and storytelling may enhance digital well-being and social connections.

In summary, our study advances emotionally intelligent VR design. Using facial expressions as the core interactions, designers can create more engaging and meaningful experiences, driving innovation in storytelling, mental health support, and virtual social interactions.

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